FEV INDEPENDENT CODING

Allowed decision values

Code	Meaning	Use when
Y	Demonstrated	The scoped record provides explicit qualifying evidence.
N	Not demonstrated	The record explicitly shows that the criterion is absent or not performed.
NR	Not reported / no qualifying evidence	The criterion cannot be supported from the reviewed sources.
U	Unclear	The record is suggestive, but the operational criterion cannot be resolved confidently.

Agentic-status rule

Full-agentic requires all three composites for the scoped use case: (A) adaptive control: F1=Y; (B) executed, stateful workflow: F3=Y and F4=Y; and (C) autonomous repair or operational completion control: F5=Y or F6=Y. **If the composite is not met, code agent-adjacent.**

Functionality dimensions

Code	Dimension	Operational question
F1	Adaptive goal-directed control	Runtime observations can select, revise, or reorder subsequent workflow actions.
F2	Coordination	Distinct agents, modules, tools, services, or human roles are coordinated within the scoped workflow.
F3	Executed actions	Tools, databases, models, code, or laboratory interfaces are executed and outputs are consumed.
F4	Retained state	Intermediate observations, files, artifacts, memory, or outputs are retained and reused downstream.
F5	Autonomous repair	A detected deficiency triggers correction, re-execution, and reuse of the corrected result.
F6	Operational decision control	The system makes a runtime accept, continue, stop, verify, or escalation decision that affects workflow progression.

Evidence dimensions

Code	Dimension	Operational question
E1	Scientific literature	Identifiable literature or passages are consumed as evidence for a runtime action or scientific claim.
E2	Structured biological knowledge	Curated databases, ontologies, annotations, pathways, knowledge graphs, or similar structured knowledge are used.
E3	Biological observations or data	Biological measurements, sequences, images, samples, variants, phenotypes, or associated metadata are analyzed.
E4	Analytical outputs	Results from executed analyses, statistics, quality control, visualizations, or intermediate computations are consumed downstream.
E5	Scientific-model outputs	Predictions, rankings, structures, designs, annotations, trajectories, or other model-generated scientific outputs are consumed downstream.
E6	Experimental or clinical observations	Experimental or clinical observations provide evidence, corroboration, validation, or feedback for the scoped scientific output.

F6 form

- Automated verification: a machine-controlled criterion or verifier governs acceptance, continuation, or completion.
- Human/higher-authority escalation: the workflow pauses or routes a decision to a user, expert, or external authority.
- Other: specify the operational mechanism and source evidence.

Cumulative Validation levels

Code	Dimension	Operational question
V0	Illustrative output	A claim, example, or generated output is shown without demonstrated end-to-end execution.
V1	Demonstrated execution	The scoped workflow is shown executing successfully.
V2	Replayable computation	Identifiable inputs, parameters, dependencies, intermediate artifacts, and execution information support reconstruction or replay.
V3	Scientific assessment	V2 is met and the agent-controlled workflow receives task-appropriate scientific assessment.

Code	Dimension	Operational question
V4	Prospective empirical testing	V3 is met and a claim-aligned output generated by the workflow is prospectively tested empirically.

Validation qualifiers

Qualifier	Meaning
B	Benchmark or comparative evaluation
H	Human or expert assessment
S	Statistical assessment
R	Robustness, ablation, failure, or sensitivity analysis
X	Independent or external assessment
P	Prospective empirical testing
C	Closed-loop empirical feedback that changes a subsequent agent-controlled action

Record 1 of 6 | GeneAgent

Record ID	GEN-012
Publication	GeneAgent: self-verification language agent for gene-set analysis using domain databases
Application category	Genomics
Scoped use case	Database-verified gene-set functional interpretation
Sources	https://www.nature.com/articles/s41592-025-02748-6 https://github.com/ncbi-nlp/GeneAgent https://zenodo.org/records/15008591

A. Agentic status

☐ Full-agentic ☐ Agent-adjacent ☐ Unclear

B. Functionality

Field	Decision
F1 **Adaptive goal-directed control**	☐ Y ☐ N ☐ NR ☐ U
F2 **Coordination**	☐ Y ☐ N ☐ NR ☐ U
F3 **Executed actions**	☐ Y ☐ N ☐ NR ☐ U
F4 **Retained state**	☐ Y ☐ N ☐ NR ☐ U
F5 **Autonomous repair**	☐ Y ☐ N ☐ NR ☐ U
F6 **Operational decision control**	☐ Y ☐ N ☐ NR ☐ U

F6 form: ☐ Automated verification ☐ Human/higher-authority escalation ☐ Other ☐ NR

C. Evidence

Field	Decision
E1 **Scientific literature**	☐ Y ☐ N ☐ NR ☐ U
E2 **Structured biological knowledge**	☐ Y ☐ N ☐ NR ☐ U

Field	Decision
E3 **Biological observations or data**	☐ Y ☐ N ☐ NR ☐ U
E4 **Analytical outputs**	☐ Y ☐ N ☐ NR ☐ U
E5 **Scientific-model outputs**	☐ Y ☐ N ☐ NR ☐ U
E6 **Experimental or clinical observations**	☐ Y ☐ N ☐ NR ☐ U

D. Validation

Cumulative level: ☐ V0 ☐ V1 ☐ V2 ☐ V3 ☐ V4

Strongest reported evaluation: ☐ Illustrative output ☐ Execution demonstration ☐ Replay assessment ☐ Retrospective scientific evaluation ☐ Prospective empirical evaluation ☐ Other

Qualifiers: ☐ B ☐ H ☐ S ☐ R ☐ X ☐ P ☐ C ☐ NR

Record 2 of 6 | SpatialAgent

Record ID	TRANS-035
Publication	SpatialAgent: An autonomous AI agent for spatial biology
Application category	Transcriptomics / spatial biology
Scoped use case	Experimental design and multimodal spatial-biology analysis
Sources	https://www.biorxiv.org/content/10.1101/2025.04.03.646459v2 https://github.com/Genentech/SpatialAgent

A. Agentic status

☐ Full-agentic ☐ Agent-adjacent ☐ Unclear

B. Functionality

Field	Decision
F1 **Adaptive goal-directed control**	☐ Y ☐ N ☐ NR ☐ U
F2 **Coordination**	☐ Y ☐ N ☐ NR ☐ U
F3 **Executed actions**	☐ Y ☐ N ☐ NR ☐ U
F4 **Retained state**	☐ Y ☐ N ☐ NR ☐ U
F5 **Autonomous repair**	☐ Y ☐ N ☐ NR ☐ U
F6 **Operational decision control**	☐ Y ☐ N ☐ NR ☐ U

F6 form: ☐ Automated verification ☐ Human/higher-authority escalation ☐ Other ☐ NR

C. Evidence

Field	Decision
E1 **Scientific literature**	☐ Y ☐ N ☐ NR ☐ U
E2 **Structured biological knowledge**	☐ Y ☐ N ☐ NR ☐ U
E3 **Biological observations or data**	☐ Y ☐ N ☐ NR ☐ U

Field	Decision
E4 **Analytical outputs**	☐ Y ☐ N ☐ NR ☐ U
E5 **Scientific-model outputs**	☐ Y ☐ N ☐ NR ☐ U
E6 **Experimental or clinical observations**	☐ Y ☐ N ☐ NR ☐ U

D. Validation

Cumulative level: ☐ V0 ☐ V1 ☐ V2 ☐ V3 ☐ V4

Strongest reported evaluation: ☐ Illustrative output ☐ Execution demonstration ☐ Replay assessment ☐ Retrospective scientific evaluation ☐ Prospective empirical evaluation ☐ Other

Qualifiers: ☐ B ☐ H ☐ S ☐ R ☐ X ☐ P ☐ C ☐ NR

Record 3 of 6 | ProtAgents

Record ID	PROT-023
Publication	Protein discovery via large language model multi-agent collaborations combining physics and machine learning
Application category	Proteomics
Scoped use case	Multi-agent protein design and analysis
Sources	https://pmc.ncbi.nlm.nih.gov/articles/PMC11235180/ https://github.com/lamm-mit/ProtAgents

A. Agentic status

☐ Full-agentic ☐ Agent-adjacent ☐ Unclear

B. Functionality

Field	Decision
F1 **Adaptive goal-directed control**	☐ Y ☐ N ☐ NR ☐ U
F2 **Coordination**	☐ Y ☐ N ☐ NR ☐ U
F3 **Executed actions**	☐ Y ☐ N ☐ NR ☐ U
F4 **Retained state**	☐ Y ☐ N ☐ NR ☐ U
F5 **Autonomous repair**	☐ Y ☐ N ☐ NR ☐ U
F6 **Operational decision control**	☐ Y ☐ N ☐ NR ☐ U

F6 form: ☐ Automated verification ☐ Human/higher-authority escalation ☐ Other ☐ NR

C. Evidence

Field	Decision
E1 **Scientific literature**	☐ Y ☐ N ☐ NR ☐ U
E2 **Structured biological knowledge**	☐ Y ☐ N ☐ NR ☐ U
E3 **Biological observations or data**	☐ Y ☐ N ☐ NR ☐ U

Field	Decision
E4 **Analytical outputs**	☐ Y ☐ N ☐ NR ☐ U
E5 **Scientific-model outputs**	☐ Y ☐ N ☐ NR ☐ U
E6 **Experimental or clinical observations**	☐ Y ☐ N ☐ NR ☐ U

D. Validation

Cumulative level: ☐ V0 ☐ V1 ☐ V2 ☐ V3 ☐ V4

Strongest reported evaluation: ☐ Illustrative output ☐ Execution demonstration ☐ Replay assessment ☐ Retrospective scientific evaluation ☐ Prospective empirical evaluation ☐ Other

Qualifiers: ☐ B ☐ H ☐ S ☐ R ☐ X ☐ P ☐ C ☐ NR

Record 4 of 6 | PathAgent

Record ID	CPBI-022
Publication	PathAgent: Toward Interpretable Analysis of Whole-slide Pathology Images via Large Language Model-based Agentic Reasoning
Application category	Computational pathology and bioimaging
Scoped use case	Multi-magnification whole-slide navigation and evidence-grounded visual question answering
Sources	https://arxiv.org/abs/2511.17052 https://github.com/G14nTDo4/PathAgent

A. Agentic status

☐ Full-agentic ☐ Agent-adjacent ☐ Unclear

B. Functionality

Field	Decision
F1 **Adaptive goal-directed control**	☐ Y ☐ N ☐ NR ☐ U
F2 **Coordination**	☐ Y ☐ N ☐ NR ☐ U
F3 **Executed actions**	☐ Y ☐ N ☐ NR ☐ U
F4 **Retained state**	☐ Y ☐ N ☐ NR ☐ U
F5 **Autonomous repair**	☐ Y ☐ N ☐ NR ☐ U
F6 **Operational decision control**	☐ Y ☐ N ☐ NR ☐ U

F6 form: ☐ Automated verification ☐ Human/higher-authority escalation ☐ Other ☐ NR

C. Evidence

Field	Decision
E1 **Scientific literature**	☐ Y ☐ N ☐ NR ☐ U
E2 **Structured biological knowledge**	☐ Y ☐ N ☐ NR ☐ U

Field	Decision
E3 **Biological observations or data**	☐ Y ☐ N ☐ NR ☐ U
E4 **Analytical outputs**	☐ Y ☐ N ☐ NR ☐ U
E5 **Scientific-model outputs**	☐ Y ☐ N ☐ NR ☐ U
E6 **Experimental or clinical observations**	☐ Y ☐ N ☐ NR ☐ U

D. Validation

Cumulative level: ☐ V0 ☐ V1 ☐ V2 ☐ V3 ☐ V4

Strongest reported evaluation: ☐ Illustrative output ☐ Execution demonstration ☐ Replay assessment ☐ Retrospective scientific evaluation ☐ Prospective empirical evaluation ☐ Other

Qualifiers: ☐ B ☐ H ☐ S ☐ R ☐ X ☐ P ☐ C ☐ NR

Record 5 of 6 | SAGA

Record ID	DRUG-012
Publication	Accelerating Scientific Discovery with Autonomous Goal-evolving Agents
Application category	Drug discovery
Scoped use case	Autonomous antibiotic discovery with deficiency-driven objective evolution and prospective antibacterial and cytotoxicity testing
Sources	https://arxiv.org/html/2512.21782v2 https://github.com/btyu/SAGA https://huggingface.co/spaces/btyu/SAGA

A. Agentic status

☐ Full-agentic ☐ Agent-adjacent ☐ Unclear

B. Functionality

Field	Decision
F1 **Adaptive goal-directed control**	☐ Y ☐ N ☐ NR ☐ U
F2 **Coordination**	☐ Y ☐ N ☐ NR ☐ U
F3 **Executed actions**	☐ Y ☐ N ☐ NR ☐ U
F4 **Retained state**	☐ Y ☐ N ☐ NR ☐ U
F5 **Autonomous repair**	☐ Y ☐ N ☐ NR ☐ U
F6 **Operational decision control**	☐ Y ☐ N ☐ NR ☐ U

F6 form: ☐ Automated verification ☐ Human/higher-authority escalation ☐ Other ☐ NR

C. Evidence

Field	Decision
E1 **Scientific literature**	☐ Y ☐ N ☐ NR ☐ U
E2 **Structured biological knowledge**	☐ Y ☐ N ☐ NR ☐ U

Field	Decision
E3 **Biological observations or data**	☐ Y ☐ N ☐ NR ☐ U
E4 **Analytical outputs**	☐ Y ☐ N ☐ NR ☐ U
E5 **Scientific-model outputs**	☐ Y ☐ N ☐ NR ☐ U
E6 **Experimental or clinical observations**	☐ Y ☐ N ☐ NR ☐ U

D. Validation

Cumulative level: ☐ V0 ☐ V1 ☐ V2 ☐ V3 ☐ V4

Strongest reported evaluation: ☐ Illustrative output ☐ Execution demonstration ☐ Replay assessment ☐ Retrospective scientific evaluation ☐ Prospective empirical evaluation ☐ Other

Qualifiers: ☐ B ☐ H ☐ S ☐ R ☐ X ☐ P ☐ C ☐ NR

Record 6 of 6 | STELLA

Record ID	CROSS-023
Publication	STELLA: Towards a Biomedical World Model with Self-Evolving Multimodal Agents
Application category	Cross-domain bioinformatics
Scoped use case	Strictosidine synthase directed evolution with prospective enzymatic testing
Sources	https://www.biorxiv.org/content/10.1101/2025.07.01.662467v2 https://github.com/zaixizhang/STELLA https://stella-agent.com/

A. Agentic status

☐ Full-agentic ☐ Agent-adjacent ☐ Unclear

B. Functionality

Field	Decision
F1 **Adaptive goal-directed control**	☐ Y ☐ N ☐ NR ☐ U
F2 **Coordination**	☐ Y ☐ N ☐ NR ☐ U
F3 **Executed actions**	☐ Y ☐ N ☐ NR ☐ U
F4 **Retained state**	☐ Y ☐ N ☐ NR ☐ U
F5 **Autonomous repair**	☐ Y ☐ N ☐ NR ☐ U
F6 **Operational decision control**	☐ Y ☐ N ☐ NR ☐ U

F6 form: ☐ Automated verification ☐ Human/higher-authority escalation ☐ Other ☐ NR

C. Evidence

Field	Decision
E1 **Scientific literature**	☐ Y ☐ N ☐ NR ☐ U
E2 **Structured biological knowledge**	☐ Y ☐ N ☐ NR ☐ U

Field	Decision
E3 **Biological observations or data**	☐ Y ☐ N ☐ NR ☐ U
E4 **Analytical outputs**	☐ Y ☐ N ☐ NR ☐ U
E5 **Scientific-model outputs**	☐ Y ☐ N ☐ NR ☐ U
E6 **Experimental or clinical observations**	☐ Y ☐ N ☐ NR ☐ U

D. Validation

Cumulative level: ☐ V0 ☐ V1 ☐ V2 ☐ V3 ☐ V4

Strongest reported evaluation: ☐ Illustrative output ☐ Execution demonstration ☐ Replay assessment ☐ Retrospective scientific evaluation ☐ Prospective empirical evaluation ☐ Other

Qualifiers: ☐ B ☐ H ☐ S ☐ R ☐ X ☐ P ☐ C ☐ NR